Principle of induction on formulas

The Open Logic Project contributors

openlogic:propositional-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted statement includes the five-case closure hypothesis (tagitem enumerate, lines 953-964); the proof delivers the author's complete smallest-class argument.

Editorial note. The label thm:induction names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Allowed standard prerequisites (editorial).

- Formulas are the smallest class containing atoms and closed under the operators. Same unit, \ollabel{defn:formulas} (formulas definition section). Lines 971-973: S satisfies the defining conditions, so Frm is contained in S.

Source reference. Original definition formulas: sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex, source lines 756–756. Notational pointer invoked inside the counted proof; it names the definition of formulas of the same unit supplying running notation, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (*Principle of induction on formulas*)**.** *If some property P holds for all the atomic formulas and is such that*

1. *it holds for $\neg A$ whenever it holds for A;*
2. *it holds for $(A \wedge B)$ whenever it holds for A and B;*
3. *it holds for $(A \vee B)$ whenever it holds for A and B;*
4. *it holds for $(A \to B)$ whenever it holds for A and B;*
5. *it holds for $(A \leftrightarrow B)$ whenever it holds for A and B;*

then P holds for all formulas.

Proof. Let S be the collection of all formulas with property P. Clearly $S \subseteq \mathrm{Frm}(\mathcal{L}_0)$. S satisfies all the conditions of formulas: it contains all atomic formulas and is closed under the operators. $\mathrm{Frm}(\mathcal{L}_0)$ is the smallest such class, so $\mathrm{Frm}(\mathcal{L}_0) \subseteq S$. So $\mathrm{Frm}(\mathcal{L}_0) = S$, and every formula has property P. $\square$